Department of Computer Science and Engineering,
Anna University, Chennai- 600025
CS23005 – Business Analytics Lab
Practical
July – November 2026
Year/Sem : IV / VII

Exercise: 03	Descriptive Data Analysis using PowerBI	24 – Jul - 2026

Observation (5 Marks)

Learn and Explore the Power BI interface

- What is Business Intelligence?
- What is Power BI?
- Differentiate Power BI Desktop and Power BI Service.
- What is Power Query?
- What is ETL?
- What is DAX?
- Difference between Calculated Column and Measure.
- What are Relationships in Power BI?
- What is a Dashboard?
- Explain Slicers and Filters.
- What is Drill Through?
- What is Forecasting?
- What are KPI Cards?
- What is Conditional Formatting?
- Explain Star Schema.

Execution (15 Marks)

Experiment: Import data for Superstore Sales Dataset (CSV), clean data, create visualizations, develop dashboards, perform business analysis, and publish reports using Power BI.

Part A – Getting Started with Power BI

- Open Power BI Desktop.
- Import the CSV dataset.
- Explore different views:
    - Report View
    - Data View
    - Model View
    - Identify
        - Number of rows
        - Number of columns
        - Data types

Part B – Data Cleaning using Power Query

- Remove duplicate records.
- Replace missing values.
- Rename columns.
- Split Full Name into
    - First Name
    - Last Name
- Change data types.
- Remove unnecessary columns.
- Apply and Close.

Part C – Sales Dashboard

- Create the following visuals:
    - Clustered Column Chart
    - Pie Chart
    - Line Chart
    - Table
    - Card Visual
- Display
    - Total Sales
    - Total Profit
    - Number of Orders
    - Top 10 Products
- Add
    - Region Slicer
    - Year Slicer

Part D – DAX Measures

- Display all measures using Cards.

Part E – Time Series Analysis

Create

- Year hierarchy
- Quarter hierarchy
- Month hierarchy

Visualizations

- Monthly Sales
- Quarterly Profit
- Year-wise Revenue
- Forecast next six months.

Part F – Customer Analytics Dashboard

- Find
  - Top Customers
  - Customer-wise Sales
  - Customer Segment
  - Average Order Value
- Create
  - Treemap
  - Funnel Chart
  - Bar Chart

Part G - Product Performance Analysis

Create visuals for

- Category Sales
- Sub-category Sales
- Profit by Product
- Top 15 Products

Use Conditional Formatting.

- Create KPI Cards
  - Highest Selling Product
  - Lowest Selling Product

Part H - Regional Sales Analysis

- Create
  - Filled Map
  - Bubble Map
  - Region-wise Sales
  - Region-wise Profit
  - State-wise Orders
- Add slicers for
  - Category
  - Region
  - Year

Part I - Executive Business Dashboard

Dashboard should include

KPIs

- Total Revenue
- Total Orders
- Total Profit
- Profit Margin

Visuals

- Line Chart
- Map
- Pie Chart
- Bar Chart
- KPI Cards

Include

- Company Logo
- Navigation Buttons
- Filters
- Drill Through
- Tooltips